

1. The denial-of-service (DoS) vector is that Moonbase selects UTXOs randomly to fund withdrawal transactions. Dishonest and wealthy users could potentially flood Moonbase with small UTXOs, making it difficult for legitimate users to find sufficiently large UTXOs to withdraw their funds.

To fix this DoS vector, Moonbase could implement a withdrawal prioritization mechanism based on the size of the UTXOs, instead of selecting UTXOs randomly. They could prioritize using the largest UTXOs available to fund withdrawal transactions. It will ensure that legitimate users with larger funds have a higher chance of successfully withdrawing their funds. It will fix the failed withdrawals because even though dishonest users flood Moonbase with numerous small UTXOs, the withdrawal prioritization mechanism ensures that legitimate users with larger UTXOs always have access to the necessary funds for withdrawal.

2. To reduce the number of UTXOs, instead of creating two separate UTXOs for each withdrawal, Moonbase can consolidate these into a single UTXO. When a user initiates a withdrawal, Moonbase can calculate the exact amount needed to fulfill the withdrawal, including any transaction fees. Moonbase can select a single UTXO from its database that is large enough to cover the withdrawal amount and the transaction fee. Moonbase can create a single withdrawal transaction that pays the target user the requested amount and sends the remaining balance back to Moonbase itself. This will simplify the UTXO arrangement process. It will reduce the number of UTXOs it needs to maintain in its database.